

Verifier-Grounded Evaluation of LLM-Generated Network Configurations: A Preregistered NetConfig-Semantic Study

Autonomous AI Researcher
CNSM 2026 Agentic AI Researcher Track

Abstract—We report a fully preregistered, artifact-anchored paired experiment (CAND-2026-001-NetConfig-Semantic-v1; preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdebb71a37) comparing a deterministic template baseline to a single-pass LLM generator (declared_model_version = gpt-5-mini) on N = 120 synthetic intent→configuration tasks using a deterministic validator. Under the frozen confirmatory semantics (shared_initial_candidate per pair) all confirmatory episodes completed: baseline = 120/120 verifier passes; guarded (gpt-5-mini, seed_0) = 120/120 verifier passes. Paired difference = 0.00; 95% bootstrap percentile CI [0.00, 0.00] (10,000 resamples, seed = 1729); exact McNemar two-sided p = 1.0. We (i) publish the exact execution and analysis artifacts and SHA-256 fingerprints needed to reproduce the run (see execution/execution_manifest.json in the artifact bundle), (ii) diagnose—using archived artifacts—why the paired outcome is uninformative about independent generative reliability (preregistered shared_initial_candidate design, template-tight task synthesis, and validator–task coupling), and (iii) provide artifact-grounded, runnable modifications (independent_per_condition sampling, multi-seed confirmatory design, validator diversity, and expanded task heterogeneity) that preserve reproducibility while producing informative independent comparisons. The immutable initial master prompt is archived (provenance/master_prompt.txt; SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdff1582bb39d15ba1).

I. INTRODUCTION

Motivation. Translating operator intent into device configuration is a core NetOps task where correctness is safety-critical: incorrect commands can break reachability, violate ACLs, or create unsafe transient states during multi-step changes. Deterministic offline verifiers (reachability/ACL/routing analyzers such as Batfish-class tools) are widely used to validate intended changes before deployment and provide an operational oracle for generated configuration artifacts [https://doi.org/10.1145/3603269.3604866]. Large language models (LLMs) offer rapid intent→config generation but raise reliability questions: how often do independently sampled LLM candidates satisfy operational verifier constraints? How do LLM pipelines compare with deterministic template baselines when correctness is judged by a deterministic validator?

Research question and contribution. After a fresh autonomous literature and gap analysis we preregistered (CAND-2026-001-NetConfig-Semantic-v1) a paired confir-

matory test asking: does a single-pass LLM generator (gpt-5-mini) have a lower verifier pass rate than a deterministic template baseline across N = 120 intent→config tasks? Our primary contribution is an artifact-anchored, fully reproducible preregistered execution + analysis bundle and a transparent diagnosis of why the preregistered confirmatory semantics (shared_initial_candidate) produced a degenerate but correct paired outcome. We (1) publish the exact artifacts and SHA-256 fingerprints required for byte-for-byte reruns; (2) report confirmatory counts and preregistered statistical inferences grounded in the archived traces (baseline = 120/120, guarded = 120/120; paired difference = 0.00; bootstrap CI [0.00,0.00]; McNemar p = 1.0); and (3) provide artifact-grounded, immediately runnable experimental modifications that preserve reproducibility while answering the operational question of independent generative reliability.

II. RELATED WORK

Verifier-grounded NetOps and intent→config. Offline semantics analyzers that compute network final/transient state from device configurations are an established safety control in NetOps; Batfish’s engineering lessons document their practical impact on pre-deployment validation and operator workflows [Matt Brown et al., 2023; https://doi.org/10.1145/3603269.3604866]. Parallel LLM→NetOps work has proposed generate–validate–repair architectures and highlighted that natural-language fluency metrics do not substitute for deterministic semantic checks; recent intent-driven generators and LLM configuration frameworks emphasize verifier grounding and human-in-the-loop guardrails [CNMS 2024; NEM 2024 — cited_record_ids].

Methodological gap this study addresses. Prior evaluations often mix generation variance, repair loops, or opaque ad-hoc scoring, and frequently omit preregistered estimands or full artifact publication. Our study contributes a frozen preregistration, deterministic scoring harness, exhaustive SHA-256 artifact manifest, and an explicit exposition of how chosen confirmatory semantics map to a specific estimand—thereby clarifying what inference the run legitimately supports and what it does not.

III. METHODOLOGY

Preregistration and frozen design. The full preregistration (CAND-2026-001-NetConfig-Semantic-v1) was archived prior to any model calls (SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fedb71a37). The preregistered confirmatory design specified a paired comparison across $N = 120$ tasks (40 tasks per topology class: edge, datacenter, multi-AS), the primary estimand `paired_success_rate_difference_guarded_minus_baseline`, confirmatory `generation_semantics` = "shared_initial_candidate", one canonical seed_0 per pair for the confirmatory analysis, the deterministic validator `deterministic_netops_validator_v5` (validator_version 5.0) and the scoring rule `full_intent_constraint_validation`. The preregistered analysis executor was `paired_binary_analysis_v1` with bootstrap inference (10,000 resamples, seed = 1729). The preregistration and analysis plan (including failure treatment, retry policy, and contamination heuristics) are published in the artifact bundle.

Task synthesis and templates. Tasks were programmatically synthesized from a deterministic template bank (`generator_id deterministic_netops_task_generator_v5`, version 5.0). Each task manifest entry encodes: the `initial_state`, `expected_final_state`, an explicit `required_sequence` (the minimal safe command sequence), per-task `workflow_policies` (e.g., `must_be_down_for_fields`, `minimum_active_in_groups`), `allowed_touched_fields`, and a difficulty annotation (`changed_interface_count`, `transient_rule_count`). These template elements convert operational constraints (make-before-break, atomic MTU changes, group availability minima) into deterministic verification criteria. Representative task manifests and their exact file locations are archived (`execution/task_manifest.jsonl`; see `artifact_examples` within the bundle).

Model, orchestration, and exact generation semantics. The declared LLM in the preregistration and execution manifest was gpt-5-mini accessed via the hosted adapter family `hosted_netops_gvr_v1`; the pinned model configuration file is `execution/model_configuration.json` (SHA-256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820). Per the preregistered confirmatory semantics we produced one canonical candidate per task (`shared_initial_candidate`) and reused that identical candidate for both baseline and guarded scoring episodes; these canonical shared candidates are archived under `execution/responses/*-shared-initial.txt` (example SHA-256s: task-000001 shared initial SHA-256 8f38c8becd7e1e36..., task-000002 SHA-256 ae967f70dfb6ccb8..., task-000003 SHA-256 f7f4db249ecbbce...). The execution manifest records that `model_calls_used` = 120 while `planned_episode_count` = 240 (the `shared_initial` design reduces distinct generation calls per condition and is explicitly documented in the manifest).

Deterministic validation and scoring. Each candidate was evaluated by `deterministic_netops_validator_v5` (validator_version 5.0). The validator pipeline performs: normal-

ization and parsing of candidate commands; enforcement of per-task `workflow_policies` and transient constraints (e.g., `minimum_active_in_groups`, `must_be_down_for_fields`); simulation of `final_state` consequences; and an intent-level binary score using the `full_intent_constraint_validation` rule. Per episode JSON scoring traces (`execution/scoring/*.json`) include `parsed_command_count`, `required_command_count`, `normalized_configuration`, `validation_before/after_final_state` dictionaries, and violation lists. Representative scoring files are archived (e.g., `execution/scoring/task-000001-guarded.json` SHA-256 0d56c1add7c5a57f...).

Execution accounting, retries, contamination heuristics. The run was executed under the `hosted_netops_gvr_v1` adapter; execution manifest entries log start/completion timestamps and exhaustive per-file SHA-256s (`execution/execution_manifest.json`). `Planned_episode_count` = 240; `completed_episode_count` = 240; `model_calls_used` = 120 (`shared_initial` generation per pair counted once); `failed_episode_count` = 0. A preregistered retry policy allowed up to two automatic retries for infrastructure failures; none were required. Contamination heuristics (exact-match and normalized n-gram overlap) were run per preregistration and recorded in `analysis/contamination_summary.csv`; no confirmatory items were flagged under the chosen thresholds (however the preregistration and Disclosure explicitly note that absence of a flag is not proof of zero pretraining exposure).

Primary inferential procedure. For each pair i ($i = 1..120$) the `baseline_i` and `guarded_i` binary verifier outcomes were recorded. The paired estimator is `mean_{i}(guarded_i - baseline_i)`. Primary inference used 10,000 bootstrap percentile replicates (seed = 1729) to form a 95% CI and employed the exact two-sided McNemar test on discordant pairs as a complementary confirmatory check. All analysis code, bootstrap parameters, and the analysis log are archived under `analysis/` (see `analysis/analysis_log.jsonl` SHA-256 ff20bc07...).

Key artifact index (representative, for rapid audit). The following canonical artifact paths and SHA-256 fingerprints are included in the submission bundle to enable byte-for-byte reruns and immediate inspection: - preregistration: `preregistration_CAND-2026-001.json` (SHA-256 7db1663cf3982c8e...) - initial master prompt: `provenance/master_prompt.txt` (SHA-256 1872df1e1805d2d9...) - model configuration: `execution/model_configuration.json` (SHA-256 a40e96e212ab87da...) - execution manifest (full artifact index): `execution/execution_manifest.json` - shared canonical candidate (example): `execution/responses/task-000001-shared-initial.txt` (SHA-256 8f38c8becd7e1e36...) - scoring trace (example): `execution/scoring/task-000001-guarded.json` (SHA-256 0d56c1add7c5a57f...) - analysis artifacts: `analysis/condition_summary.csv` (SHA-256 9c77c96f37c4b6ff...), `analysis/paired_contingency_table.csv` (SHA-256 9f25790c7eeafb3...).

Deviations and transparency. The methodology above follows the frozen preregistration; all deviations, retries, exclusions, and per-file SHA-256 fingerprints are recorded in

the execution manifest and analysis logs. The artifact bundle provides the exact provenance needed for external auditors to re-execute the analysis under alternative generation semantics (e.g., `independent_per_condition` or `multi-seed confirmatory` designs) without introducing unspecified choices.

IV. RESULTS

Confirmatory outcomes (artifact-anchored). All preregistered confirmatory scoring episodes completed. The archived confirmatory counts are: `baseline_success_count` = 120/120; `guarded_success_count` = 120/120; `complete_pair_count` = 120. The paired contingency table is degenerate: `n_11` = 120, `n_10` = 0, `n_01` = 0, `n_00` = 0 (see `analysis/paired_contingency_table.csv`; SHA-256 9f25790c...). The preregistered paired estimator (mean of `guarded_i` - `baseline_i`) = 0.00. Bootstrap inference (10,000 resamples; seed = 1729) yields a 95% percentile CI [0.00, 0.00]; exact McNemar two-sided p = 1.0. All analysis artifacts (`condition_summary.csv` SHA-256 9c77c96f..., `paired_difference_figure.svg` SHA-256 ae5b8e0c...) are archived for byte-for-byte verification.

Why the contingency is degenerate (artifact diagnosis). Two preregistered, observable design choices (both visible in the artifact bundle) explain the degenerate result. First, `confirmatory generation_semantics` = "shared_initial_candidate" intentionally reuses a single canonical candidate per task for both the baseline and guarded scoring episodes; those canonical candidates are archived at `execution/responses/task-000XXX-shared-initial.txt` (representative SHA-256s: `task-000001` 8f38c8be..., `task-000002` ae967f70..., `task-000003` f7f4db24...). Second, the task templates encode explicit `required_sequences` and transient `workflow_policies` (e.g., `must_be_down_for_fields`, `minimum_active_in_groups`) that the canonical candidates satisfy. Because `deterministic_netops_validator_v5` deterministically maps the identical canonical input to the same pass/fail outcome, the paired counts are necessarily identical (`n_11` for all pairs). The scoring traces make this chain of logic directly auditable: e.g., `execution/scoring/task-000001-guarded.json` (SHA-256 0d56c1ad...) shows `parsed_command_count` = 4, `required_command_count` = 4, `violation_count` = 0 and contains before/after `final_state` dictionaries; the corresponding `shared_initial` candidate file `execution/responses/task-000001-shared-initial.txt` (SHA-256 8f38c8be...) matches the candidate text used for both conditions.

Validator trace evidence (representative examples). Per-episode JSON scorer outputs include validator fields used in our interpretation: `normalized_configuration`, `parsed_command_count`, `required_command_count`, `validation_before.final_state` and `validation_after.final_state`, `violation_count`, and `scorer_id/version`. Example artifacts (all archived): - `execution/scoring/task-000001-guarded.json` (SHA-256 0d56c1ad...): `parsed_command_count`=4; `required_command_count`=4; `valid`=true; `final_state` mapping

present. Shared candidate: `execution/responses/task-000001-shared-initial.txt` (SHA-256 8f38c8be...). - `execution/scoring/task-000002-guarded.json` (SHA-256 993b8eb3...); shared candidate: `execution/responses/task-000002-shared-initial.txt` (SHA-256 ae967f70...). - `execution/scoring/task-000003-guarded.json` (SHA-256 dad1cf4c...); shared candidate: `execution/responses/task-000003-shared-initial.txt` (SHA-256 f7f4db24...). These traces demonstrate that the validator confirmed required sequences and transient constraints for the canonical candidates; full JSON traces are available under `execution/scoring/` for audit.

Provider-call and execution accounting evidence. The execution manifest records provider call traces and caching (`execution/provider_calls/*.json` and `execution/call_cache/*`). Key archived counts: `planned_episode_count`=240, `completed_episode_count`=240, `model_calls_used`=120, `maximum_model_calls`=240. No failed provider calls occurred; the preregistered retry policy permitted up to two retries for infrastructure failures but none were invoked. The exhaustive execution artifact manifest (`execution/execution_manifest.json`) contains per-file SHA-256 fingerprints for deterministic reproduction.

Contamination checks and caveats. The preregistered contamination heuristics (exact match and normalized n-gram overlap thresholds) produced no confirmatory flags (`analysis/contamination_summary.csv` is empty under the preregistered thresholds). As stated in the preregistration and Disclosure, the absence of a flag is not proof of zero pretraining exposure; full provider call traces and `call_cache` artifacts are archived to permit more intensive exposure analyses by third parties.

Implication of the numeric results. The confirmatory statistics are internally correct and properly reported for the preregistered estimand: they test whether a deterministic verifier treats identical artifacts differently across conditions. They do not estimate the operational quantity of independent LLM generative reliability (the probability that independently sampled LLM candidates satisfy verifier constraints). The archived artifacts both demonstrate why (shared canonical inputs) and provide all materials necessary to run the alternate, informative protocols (`independent_per_condition` sampling, `multi-seed confirmatory` execution, `validator diversification`) without any nondeterministic gaps in reproducibility.

V. DISCUSSION

Interpretation of confirmatory inference. The confirmatory analysis correctly reports that, for the preregistered estimand and generation semantics, the deterministic verifier returned identical pass judgments for both conditions: `baseline_success_count` = 120/120 and `guarded_success_count` = 120/120, producing a paired difference of 0.00 with a degenerate 95% bootstrap percentile CI [0.00,0.00] (10,000 resamples, seed = 1729) and exact McNemar p = 1.0. This numerical outcome is a direct and reproducible consequence of two archived facts: (A) the preregistered `generation_semantics` = "shared_initial_candidate" produced one canonical

candidate per task that was reused for both the baseline and guarded episodes (see `execution/responses/*-shared-initial.txt`; representative SHA-256s include `task-000001: 8f38c8becd7e1e36...`, `task-000002: ae967f70dfb6ccb8...`, `task-000003: f7f4db249ecbbce...`), and (B) the deterministic validator (`deterministic_netops_validator_v5`) is idempotent on identical inputs and enforces the template-encoded `required_sequences` and transient safety constraints exactly (see `execution/scoring/*.json`; representative SHA-256 `execution/scoring/task-000001-guarded.json 0d56c1add7c5a57f...`). Because both conditions received byte-identical canonical candidates, identical validator outputs were the only logically possible result; the archived `execution_manifest` records `model_calls_used = 120` while `planned_episode_count = 240`, confirming the single-generation per pair accounting (`execution/execution_manifest.json` SHA-256 `7db1663cf...` in preregistration and manifest entries).

What this confirms—and what it does not. The run validates the preregistered estimand (whether the verifier treats identical inputs differently across scoring conditions) and demonstrates complete reproducibility through the published artifact manifest and per-file SHA-256s. It does not, however, estimate independent LLM generative reliability (the per-task probability that an independently sampled LLM candidate satisfies the verifier). That latter operational quantity requires independent per-condition sampling (distinct generation seeds or independent generation calls per condition), which the archived manifest shows was not performed for the confirmatory test (`model_calls_used = 120` vs. the 240 episodes scored). Interpreting the observed 120/120 pass rates as evidence that independently sampled LLM outputs will pass the validator at the same rate would therefore be a category error; the artifacts make the provenance of this limitation transparent.

How archived artifacts enable immediate, rigorous reruns. One virtue of full artifact publication is that the minimal changes needed to answer the operational question are runnable and deterministic. Using the provided bundle (`execution/responses`, `execution/provider_calls`, `execution/call_cache`, `execution/scoring`, `execution/task_manifest.jsonl`, and `execution/model_configuration.json`), researchers or practitioners can (without changing task templates or the validator) (i) switch to `independent_per_condition` semantics (generate a distinct candidate per condition per task) and reexecute scoring; this deterministically increases `model_calls_used` from 120 toward the preregistered maximum of 240 and breaks the shared-input coupling that produced the degenerate contingency, or (ii) adopt a multi-seed confirmatory design ($K \geq 3$ independent seeds per condition) to estimate per-condition pass probabilities with bootstrap CIs. Both reruns preserve byte-for-byte reproducibility for all unchanged artifacts because the original provider traces and call cache are published (`execution/provider_calls/*.json` and `execution/call_cache/*`). The manifest documents exact file paths and SHA-256s so external auditors can verify that only the generation semantics changed while all other artifacts remain identical.

Validator diversity and construct validity. The deterministic validator enforces the template-expressed `workflow_policies` exactly; this is an asset for operational safety but a construct-validity risk for generalization because template-aligned canonical candidates are more likely to match validator expectations. The archived scoring traces quantify this coupling (`parsed_command_count`, `required_command_count`, and the `validation_before/validation_after` `final_state` in `execution/scoring/*.json`). A pragmatic mitigation—supported by the archived bundle—is to add a second independent validator (e.g., a reachability emulator or an alternate static analyzer) and report both intersection and union pass rates. Because validator outputs are archived as JSON traces, adding a validator is an orthogonal rerun that preserves the original artifacts while increasing construct validity of the decision rule.

Threats to validity and how they are addressed by the artifacts. Internal validity: the `shared_initial` design intentionally removed generation variance and thus precluded testing independent generative reliability; this is visible in the manifest (`model_calls_used = 120`) and in the repeated `shared_initial` SHA-256 fingerprints. Construct validity: the task templates' explicit `required_sequences` and `workflow_policies` increase the probability that a canonical, template-aligned candidate will pass the validator; the task manifest entries (`execution/task_manifest.jsonl`) and per-task payloads in the bundle document these constraints and make the construct limitation auditable. External validity: the study used a single declared model (gpt-5-mini) and synthetic, template-generated tasks; all such scope limitations are preserved in the preregistration and manifest so consumers can judge applicability. Conclusion validity: the degenerate contingency yields exact but uninformative inference about independent generative reliability; the archived bootstrap parameters (10,000 resamples, `seed = 1729`) and generated bootstrap artifact (`analysis/paired_difference_figure.svg`) ensure this conclusion is verifiable.

Operational implications for NetOps. From an operator perspective the artifact-anchored diagnosis yields three practical rules grounded in the archived evidence: (1) Verifier grounding is necessary but insufficient—experimental generation semantics must match the deployment question. The manifest proves that identical inputs produce identical verifier outputs; reuse of canonical candidates therefore invalidates claims about independent sampling. (2) Reproducible preregistration + artifact publication materially reduces ambiguity about what a confirmatory test measures and accelerates corrective reruns that answer operational questions (the exact rerun steps are fully executable from the published bundle). (3) For deployment decisions, practitioners should report per-condition pass probabilities estimated under independent sampling and, when possible, validate across multiple independent validators; both methodological changes are artifact-supported reruns rather than new experiments and preserve full reproducibility.

Concluding remark. The confirmatory run executed exactly

as preregistered and the archive clearly documents why the observed paired outcome is degenerate for the operational quantity of independent LLM reliability. That transparent, artifact-level diagnosis is the central scientific value of this submission: it shows how a carefully preregistered, fully archived pipeline can expose subtle estimation/semantics mismatches and enable deterministic, reproducible corrective reruns that answer the operational questions NetOps teams require.

VI. LIMITATIONS

- Controlled synthetic tasks: programmatic templates produced narrow required_sequences and workflow_policies that favour canonical solutions and limit external generalizability to heterogeneous operational intents.
- Confirmatory shared_initial semantics: the preregistered generation_semantics = 'shared_initial_candidate' supplied identical canonical candidates to both conditions per pair, reducing independence between conditions and causing the degenerate paired outcome.
- Single canonical seed for confirmatory test: the primary analysis used one canonical seed per task; preregistered exploratory multi-seed analyses (seeds_1..4) were not included in the confirmatory inference reported here.
- Validator-task coupling: the deterministic validator enforces constraints encoded in the task templates, which can increase pass rates for template-aligned candidates and reduce construct validity for open distributions.
- Possible training-data exposure: preregistered contamination heuristics found no flags under chosen thresholds, but absence of a flag is not proof of zero exposure to related artifacts in model pretraining.
- Single-model, single-validator run: results apply only to gpt-5-mini under the recorded configuration and deterministic_netops_validator_v5; they do not generalize across model families, agentic pipelines, or other validators.

VII. CONCLUSION

We executed a fully preregistered, verifier-grounded paired experiment and released a complete artifact bundle enabling byte-level reproduction (execution/execution_manifest.json and analysis/*). Under the preregistered confirmatory semantics (shared_initial_candidate) both the deterministic template baseline and the gpt-5-mini guarded condition validated on all N = 120 tasks (both 120/120). Archived evidence (shared_initial candidate files and validator scoring traces) demonstrates that identical canonical candidates per pair and template-tight task constraints caused the degenerate paired outcome; consequently the confirmatory paired result does not provide evidence about independent LLM generative reliability in operational settings. We provide artifact-grounded, runnable modifications (independent_per_condition sampling, multi-seed confirmatory execution, validator diversification, task heterogeneity expansion) that preserve reproducibility and

enable informative independent comparisons. All execution and analysis artifacts—including the immutable master prompt reference (provenance/master_prompt.txt; SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15) and the preregistration SHA-256—are included in the submission bundle to permit exact audit and follow-up experiments. Transparent preregistration combined with exhaustive artifact publication clarifies precisely what a confirmatory test measures and accelerates corrective reruns that answer the operational questions NetOps practitioners require for deployment decisions.

DISCLOSURE STATEMENT

Mandatory disclosure (CNSM Agentic AI Researcher track). LLM(s) and provider: declared_model_version = gpt-5-mini accessed via provider adapter 'openai_responses' and adapter family 'hosted_netops_gvr_v1' (execution/model_configuration.json SHA-256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd82). Orchestration/framework: hosted_netops_gvr_v1 executed the preregistered pipeline; deterministic scoring used deterministic_netops_validator_v5 (validator_version 5.0). Preregistration: CAND-2026-001-NetConfig-Semantic-v1 (frozen prior to execution); preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdded71a3. Exact initial master prompt: preserved as an immutable archived file provenance/master_prompt.txt (SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15). Human role and autonomy boundary: the Research Director selected the topic family (Generative AI and LLMs for NetOps) and approved the initial master prompt before the autonomy lock. After the autonomy lock the autonomous pipeline performed literature review, research-gap selection, preregistration, code generation, experiment execution, deterministic validation, statistical analysis, autonomous internal peer review, manuscript drafting and revision. No human manually edited technical manuscript content after the autonomy lock. Preregistered confirmatory generation semantics and consequence: confirmatory generation_semantics was set to 'shared_initial_candidate' so each pair received an identical canonical candidate for both baseline and guarded episodes; representative shared candidate artifacts: execution/responses/task-000001-shared-initial.txt (SHA-256 8f38c8becd7e1e36...), execution/responses/task-000002-shared-initial.txt (SHA-256 ae967f70dfb6c...), execution/responses/task-000003-shared-initial.txt (SHA-256 f7f4db249ecbbce...). Validator and scoring: deterministic_netops_validator_v5 performed normalized parsing, intent constraint checking, transient safety checks, and emitted per-episode JSON scoring traces archived under execution/scoring/*_json (representative SHA-256: execution/scoring/task-000001-guarded.json SHA-256 0d56c1add7c5a57f...). Execution accounting and retries: planned_episode_count = 240, completed_episode_count = 240, model_calls_used = 120, failed_episode_count = 0; preregistered retry policy allowed up to 2 automatic

retries for infrastructure failures; none were required. Contamination checks and reproducibility: preregistered string-overlap heuristics found no confirmatory flags under chosen thresholds, but absence of a flag is not proof of zero exposure to related artifacts in model pretraining. All artifacts, seeds, code, and per-file SHA-256 hashes required for exact reproduction are released in the submission bundle (execution/execution_manifest.json contains the exhaustive manifest). The initial master prompt is referenced immutably by path and SHA-256 above.

REFERENCES

- [1] <https://doi.org/10.1145/3603269.3604866>
- [2] <https://doi.org/10.23919/cnsm62983.2024.10814407>
- [3] <https://doi.org/10.1002/nem.70029>